Arnav Gupta

Independent ML Researcher | Bandipur, Siraha, Nepal

arnav.gupta.ai@outlook.com | ORCID | Google Scholar | GitHub | X | LinkedIn | Substack | arnavgupta.com.np

About

Independent ML researcher whose work is animated by a structural question: when does the spectral and geometric character of a learned representation determine what a learning system can and cannot do—and what does that imply about when pipelines will fail? Two preprints pursue this from complementary angles. The first (arXiv, 2026, v2) introduces the spectral saturation index $S(K) = \mathrm{erank}(\hat{\Sigma}_W^{(K)})/K$, a label-free stopping rule for few-shot label acquisition validated across 49 real tasks and three frozen backbones. The second (Research Square, 2026) provides the first controlled empirical documentation of verifier exploitation in NLI-guided iterative refinement—a failure mode in which a system learns to game its own verifier rather than improve factual faithfulness. Both investigations were conducted without institutional affiliation, advisor, or supervised compute, from Nepal, at fifteen.

Publications & Preprints

The Geometry of Saturation: Effective Rank Predicts When Labels Stop Helping in Few-Shot Classification

Arnav Gupta. *Independent Researcher, Nepal.* arXiv preprint, 2026 (v2). `doi:10.48550/arXiv.2606.24903`

- Introduced the **spectral saturation index** $S(K) = \mathrm{erank}(\hat{\Sigma}_W^{(K)})/K$—a closed-form, label-free scalar grounded in classical high-dimensional covariance estimation (Roy & Vetterli, 2007)—that predicts when labeling additional support examples yields diminishing returns, computable in $O(d^3)$ time with no test labels and no trained classifier.
- Validated across **49 real tasks** spanning binary, 5-way, and 10-way classification on **3 frozen backbones** (PCA-50, CLIP ViT-B/32, DINOv2 ViT-S/14) and 6 datasets; **492 doubling-pair observations**.
- Pooled Spearman $\rho = 0.6366$ ($p = 2.9 \times 10^{-57}$, $N = 492$); median within-task $\rho = 0.767$; cluster-bootstrap 95% CI $[0.551, 0.720]$. Partial Spearman controlling for $\log K$ yields $\rho_{\text{partial}} = 0.324$ ($p = 1.65 \times 10^{-13}$), confirming spectral information beyond shared K-dependence.
- Fixed threshold $\tau = 0.02$ classifies stop/continue decisions with **cluster-bootstrap AUC** = 0.787 $[0.713, 0.860]$, achieving **100% recall on meaningful gains** ($\Delta A > 1\%$). Theory proves $K_{\text{sat}} \approx \mathrm{erank}(\Sigma_W)/\tau$ with high probability under sub-Gaussian class-conditionals.
- Two-regime practical rule: **PCA-50** (hard stop— halt when $S(K) < 0.02$, saturation within practical K); **CLIP/DINOv2** (diminishing-returns monitor—watch $S(K)$ drop from $\sim 0.3 \to 0.05$, saturation beyond practical label budgets). Computation ~ 1 ms on CPU at $d = 50$.

Verifier Exploitation in NLI-Guided Iterative Refinement: A Controlled Empirical Analysis

Arnav Gupta. *Independent Researcher, Nepal.* Research Square preprint, 2026. `doi:10.21203/rs.3.rs-10187492/v1`

- First controlled empirical documentation of verifier exploitation in zero-gradient, prompt-only iterative refinement: a second revision cycle inflated SummaCConv by +0.185 while collapsing $\text{BARTScore}_{s\to d}$ by -2.566 nats, confirmed by Wilcoxon $W = 0$ ($p = 2.68 \times 10^{-83}$) across all 498 instances without exception.
- Identified *truncation-exploiting content removal* as the exploitation mechanism; formalized a three-condition, annotation-free detection protocol applicable to any NLI-guided iterative refinement pipeline.
- AnchorSum achieves a 6.3% relative SummaCConv inconsistency reduction over 498 Multi-News instances ($p = 4.49 \times 10^{-28}$); outperforms fine-tuned baselines (BART, PEGASUS, PRIMERA) across all LLM-as-judge dimensions in a dual-judge jury evaluation.

Education

Sagarmatha Secondary School — Mirchaiya, Siraha, Nepal
Grade 10 *GPA: 3.87/4.00* (Outstanding) — *Expected SEE: 2027*

Experience

Suvidha Foundation — September 2025 – November 2025
Machine Learning Intern — *Remote*
- Researched multi-document abstractive summarization with transformer-based models under mentorship of a PhD in NLP.
- Built a proof-of-concept summarization system incorporating hierarchical context aggregation across multiple source documents.
- Participated in community outreach programs supporting underserved populations alongside technical work.

Programs & Fellowships

Harvard Computer Society AI Bootcamp — 2025
Artificial Intelligence Fellow — *Remote*
- Intensive AI fellowship with Harvard students focused on frontier AI systems, modern architectures, and real-world deployment challenges.
- Implemented a Transformer encoder from scratch: sinusoidal positional encoding, multi-head self-attention, cross-attention, and position-wise feed-forward networks.
- Surveyed RL foundations (MDPs, Value/Policy Iteration, SARSA, Q-Learning, Multi-Armed Bandits) and frontier LLM architectures with focus on hallucination mitigation and inference-time reliability.
- Completed hands-on projects: fine-tuned BERT, built a mini-LLM, implemented Vision Transformers, constructed neural networks from first principles.

Projects

NietzscheGPT *Python, PyTorch, Hugging Face Transformers* `[Hugging Face]` — 2026
- Fine-tuned GPT-2 Medium (355M parameters) on the complete English works of Nietzsche (25 books, ≈2.2M tokens) to generate philosophical text in his style; trained on Apple M3 Pro via MPS backend in ≈4 hours and published on Hugging Face.

Transformer from Scratch *Python, PyTorch, NumPy* `[GitHub]` — 2025
- Full Transformer encoder from first principles—sinusoidal positional encoding, multi-head self-attention, position-wise FFN, residual connections, layer normalization—trained on a synthetic autoregressive task; modular notebook bridging mathematical derivations and working code.

10D Hypercube Visualization *Python, NumPy, Scikit-learn, Matplotlib* [GitHub] 2026

— Constructed and visualized a 10-dimensional hypercube (1,024 vertices; 5,120 edges) using PCA, t-SNE, and layer-stratified views; generated comparative panels illustrating how geometric structure is preserved or distorted under each projection.

House Price Prediction *Python, Scikit-learn, XGBoost, Pandas* [GitHub] 2025

— Regression pipeline on Ames Housing (Kaggle): EDA, feature engineering (composite features, log transforms, ordinal encoding); tuned XGBoost (CV RMSE_{log} 0.1229) outperforming Ridge baseline (0.1445) via GridSearchCV.

Live Emotion Detection *Python, DeepFace, OpenCV, TensorFlow, MTCNN* [GitHub] 2025

— Real-time 7-class facial emotion classifier using DeepFace with MTCNN detection; temporal smoothing via 10-frame sliding window, frame-skipping for throughput, overlaid confidence bars and FPS counter.

Technical Skills

Languages	:	Python, C++, SQL
Libraries & Frameworks	:	PyTorch, TensorFlow, Hugging Face Transformers, Scikit-learn, NumPy, Pandas, Matplotlib, XGBoost, OpenCV, DeepFace
Tools & Platforms	:	Git/GitHub, Docker, Jupyter, VS Code, CUDA, Firebase, MySQL, REST APIs, Zotero, Obsidian
Typesetting	:	LaTeX, Markdown

Mathematics Coursework (MIT OpenCourseWare)

— **18.01 — Single Variable Calculus** (Completed): Differential and integral calculus with rigorous problem-solving.

— **18.06 — Linear Algebra** (In Progress): Matrix theory, eigendecomposition, orthogonality, and applications to machine learning.